

Proposal: Extending the University Guidance System for Multi-Department Operation

Prepared for: Guidance Department Leadership / Project Stakeholders

Status: For Review (Revision 5 — Complete Integrated Architecture, BPM & UI Roadmap)

Supersedes: Monolithic Architecture Proposal (v1) & Revisions 2–4

1. Purpose

The current platform was designed as an isolated web application for the Guidance Department[cite: 1, 2]. Other campus departments (FacultyPortal, Finance, Library, Registrar, StudentPortal) are adopting the same core architecture[cite: 1, 2].

This proposal defines the minimum set of changes required to enable cross-department operation and inter-service collaboration while respecting the physical isolation of the six independent ASP.NET Core department projects (FacultyPortalMain, FinanceMain, GuidanceDepartmentMain, LibraryMain, RegistrarMain, StudentPortalMain)[cite: 1, 2]. It establishes an inter-service referral protocol and cross-app trust model without compromising student privacy, workflow integrity, or local project boundaries[cite: 1, 2].

2. Background

The workspace consists of six independent ASP.NET Core (net10.0) web applications running on separate high localhost ports (52141–52146)[cite: 1, 2]. Shared capabilities (Authentication, Request Management, Triage, Audit Logging, PII Masking, CSV Import, and Polly Notifications) originate in GuidanceDepartmentMain and are copied to target projects using the Install-GuidanceServices.ps1 installer script[cite: 1, 2].

Existing Strengths

- Independent runtime, process, and data store isolation per department[cite: 1, 2].
- Role-based access control ([Authorize], Student/Counselor roles)[cite: 1, 2].
- Ownership enforcement scoped to StudentId[cite: 1, 2].
- Audit logging of local state transitions (Requested -> Resolved)[cite: 1, 2].
- PII masking on export and strict CSV import validation[cite: 1, 2].
- Polly circuit breaking and retry handling in NotificationService[cite: 1, 2].

Current Architectural Gaps

- Isolated Silos: Each department app operates as an isolated silo with no mechanism for inter-service communication or cross-app trust[cite: 1, 2].

- Missing Inter-Service Referral Protocol: Counselors cannot refer requests to another department. The existing two-state model only supports local processing[cite: 1, 2].
- UI-Only Department Placeholders: All department-facing UI pages remain presentation-only placeholders driven by prototype.js[cite: 1, 2].

3. Problem Statement

The current system lacks an Inter-Service Protocol[cite: 1, 2]. Because each department runs as an independent web application with its own isolated store, cross-department operations cannot be solved by simply adding a DepartmentId column to a single database schema[cite: 1, 2].

Without an explicit inter-service protocol:

- Staff cannot forward or refer student requests to another department application[cite: 1, 2].
- Cross-department referrals would require manual, out-of-band communication, breaking audit trails[cite: 1, 2].
- Copying data manually between applications creates risks of PII leakage and data desynchronization[cite: 1, 2].

4. Proposed Solution

4.1 Department Identity & Cross-App Trust Architecture

Rather than introducing a single multi-tenant database, each department project will maintain an explicit Department Identity and trust configuration[cite: 1, 2]:

- Local Department Identity: Each appsettings.json specifies its code and local endpoints[cite: 1, 2].
- Service-to-Service Trust: Standardize M2M authentication via shared JWT signing keys or API keys across ports 52141–52146[cite: 1, 2].
- Service Layer Propagation: Shared contracts originate in GuidanceDepartmentMain and sync via Install-GuidanceServices.ps1[cite: 1, 2].

4.2 Cross-Department Referral Protocol

- Local State Transitions: Extend workflow from Requested -> Resolved to include ReferredOut and ReferredIn / Requested[cite: 1, 2].
- Inter-Service REST Endpoint: Expose POST /api/v1/incoming-referral across all departments[cite: 1, 2].
- Data Minimization (Safety Valve): Referrals send minimal payload (StudentId, sanitized referral reason, originating reference ID). Full case notes stay local[cite: 1, 2].
- Resilient Outbound Delivery: Uses Polly retries and circuit breaking in NotificationService[cite: 1, 2].
- Audit Event Capture: Both ReferralSent and ReferralReceived write to local AuditLogService[cite: 1, 2].

5. Guidance Department Business Process Model (BPM)

Department: Guidance Department (Source / Canonical Implementation)
Project Path: Departments/GuidanceDepartment/GuidanceDepartmentMain
Port: 52143 | Target Framework: net10.0

5.1 Department Actors & Roles

Actor / Component	Role	Boundary & Access
Student	Guidance client seeking support	Accesses request.html via claims-derived identity (Student role).
Guidance Counselor	Case handler & triage staff	Accesses triage.html (Counselor role) to review queues and case notes.
Department Admin	System lead / Department head	Accesses admin.html for local staff routing and roster imports.
Guidance Core Engine	Local service & business rule enforcement	Port 52143. Handles JWT verification, sanitization, and PII masking.
External Endpoints	Receiving/Referring departments	Ports 52141–52146. Communicates via M2M REST payloads.

5.6 Guidance Request State Machine Definitions

Status	Trigger	Description	Data Retained
Requested	Student Submission	Case pending initial counselor review.	Full submission + Student Claims
ReferredIn	Inbound M2M Call	Ingested from external department. Prioritized in queue.	StudentId, ReferralReason, OriginatingDeptCode
InReview	Counselor Action	Counselor actively conducting triage or session.	Local confidential counseling notes
ReferredOut	Counselor Referral	Request transferred to another department.	Sanitized reference pointer, dispatch audit record
Resolved	Counselor Resolution	Case closed with outcome summary.	Resolution summary, immutable audit log entry

6. Scope

In Scope: DepartmentIdentity block, ReferredOut/ReferredIn states, ReferRequestDto, POST /api/v1/incoming-referral endpoint, Polly transport integration, and static UI migration roadmap[cite: 1, 2].

Out of Scope: Monolithic database merging, JWT cookie session changes, FluentValidation patterns, and mobile PWA engine[cite: 1, 2].

7. Implementation Approach & UI Roadmap

7.2 Central AI Context Synchronization Plan

Section to Update	Required Change
Shared Service Layer	Add ReferralService and InterServiceAuthHandler to list of shared services[cite: 1, 2].
Contracts	Document ReferRequestDto and ServiceResult additions in Contracts/ServiceContracts.cs[cite: 1, 2].
Dependency Registration	Update Program.cs to include scoped registrations for inter-service referral handlers[cite: 1, 2].
Security Boundaries	Add rules for M2M authorization keys or shared JWT signing keys across ports[cite: 1, 2].

7.3 Static Page UI Roadmap

Static Page Route	Current State	Target Backend Capability	Transition Phase
/request.html	Presentation placeholder	Connects submission to StudentRequestService[cite: 2].	Phase B
/triage.html	Presentation placeholder	Displays live queues including ReferredOut/ReferredIn lanes[cite: 2].	Phase B
/monitoring.html	Mock register	Connects to PiiMaskingService and live registers[cite: 2].	Phase C

/dashboard.html	Mock metrics	Queries live counts of active and referred requests[cite: 2].	Phase D
/admin.html	Settings preview	Upgraded to manage M2M trust and staff routing rules[cite: 1, 2].	Phase D
/case-notes.html	UI placeholder	Strictly local to originating department; protected by auth[cite: 2].	Phase D / Local
/appointments.html	UI placeholder	Remains local UI mock until API contracts are scoped[cite: 2].	Future Roadmap
/resources.html	Static resources	Pure static presentation content; no backend API required[cite: 2].	No Change

8. Risks and Considerations

1. Target Service Unavailability: Mitigated via Polly retries and circuit breakers in NotificationService[cite: 1, 2].
2. Data Over-Sharing: Mitigated by strict ReferRequestDto data minimization[cite: 1, 2].
3. Service Drift: Mitigated by maintaining GuidanceDepartmentMain as canonical source and running Check-GuidanceServices.ps1[cite: 1, 2].

9. Recommendation & Action Plan

Item / Deliverable	Type	Effort	Status
Phase A: Department Identity & Service Synchronization	Core Infrastructure	Medium	Not Approved[cite: 1, 2]
Phase B: Inter-Service Referral Protocol & Core UI	New Workflow	Medium	Not Approved[cite: 1, 2]
Phase C: Department-Aware PII Masking & Monitoring	Service Enhancement	Low-Medium	Not Approved[cite: 1, 2]
Phase D: Department Admin & Dashboard Integration	UI / Config Extension	Low-Medium	Not Approved[cite: 1, 2]

Design Variance Protocol (Section 9.2)	Pre-Dev Process	N/A	Adopted (Revision 5)[cite: 2]
Central AI Context Sync Plan (Section 7.2)	Documentation	N/A	Adopted (Revision 5)[cite: 1, 2]
Static Page UI Roadmap (Section 7.3)	UI Architecture	N/A	Adopted (Revision 5)